

LUHANKA JUDINSALO, Finland

WMO: 027650

Lat: 61.704N

Lon: 25.505E

Elev: 83

StdP: 100.33

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -24.1	(c) -20.2	(d) -26.3	(e) 0.3	(f) -24.0	(g) -22.1	(h) 0.5	(i) -19.5	(j) 14.6	(k) -3.2	(l) 13.0	(m) -3.9	(n) 2.2	(o) 10	(p) 0.667

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.1	26.1	19.3	24.0	18.3	22.3	17.4	20.5	24.3	19.4	23.0	18.2	21.5	3.0	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.1	14.0	22.7	18.0	13.0	21.8	16.7	12.0	20.4	59.3	24.1	55.9	23.2	51.7	21.5	23.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.1	10.6	9.4	-24.7	27.2	6.5	2.8	-29.4	29.1	-33.2	30.8	-36.8	32.3	-41.5	34.3
			-24.9	21.0	6.5	1.4	-29.5	22.0	-33.3	22.9	-36.9	23.7	-41.6	24.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	5.4	-6.5	-5.2	-2.5	2.6	9.8	14.4	18.0	16.4	12.0	5.7	1.6	-2.1	(6)
(7)		DBStd	9.33	6.35	6.49	4.55	3.52	4.01	3.38	3.49	2.75	2.91	3.29	4.08	5.49	(7)
(8)		HDD10.0	2382	512	426	387	222	54	3	0	0	16	137	252	374	(8)
(9)		HDD18.3	4777	770	659	646	472	265	124	51	73	192	392	502	632	(9)
(10)		CDD10.0	707	0	0	0	1	47	137	248	197	74	3	0	0	(10)
(11)		CDD18.3	61	0	0	0	0	1	7	41	12	0	0	0	0	(11)
(12)		CDH23.3	247	0	0	0	0	2	33	181	31	0	0	0	0	(12)
(13)		CDH26.7	34	0	0	0	0	0	4	27	3	0	0	0	0	(13)
(14)	Wind	WSAvg	4.2	4.3	4.3	4.4	3.7	3.7	4.1	3.7	4.0	4.2	4.9	4.8	5.0	(14)
(15)	Precipitation	PrecAvg	618	48	35	31	33	41	63	82	70	52	62	53	49	(15)
(16)		PrecMax	807	89	79	62	58	107	103	165	137	140	131	113	94	(16)
(17)		PrecMin	510	11	2	3	6	8	27	16	15	9	14	4	13	(17)
(18)		PrecStd	74	21	19	13	16	23	21	36	31	29	28	24	20	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.7	5.4	7.8	15.8	23.0	27.0	29.2	26.9	19.7	13.0	10.8	7.7	(19)
(20)			MCWB	2.8	2.9	3.8	9.1	13.7	16.3	20.5	20.6	15.9	11.3	9.2	7.0	(20)
(21)		2%	DB	2.4	3.4	5.4	11.6	20.6	23.6	27.0	23.7	18.0	12.0	8.8	5.4	(21)
(22)			MCWB	1.9	2.2	2.7	6.6	13.4	16.7	19.6	18.7	14.9	10.7	8.0	4.5	(22)
(23)		5%	DB	1.5	2.0	3.9	9.5	18.4	21.5	25.4	21.9	16.8	11.2	7.6	4.2	(23)
(24)			MCWB	1.1	1.3	1.5	5.5	12.5	15.3	19.3	17.3	14.3	10.0	6.9	3.4	(24)
(25)		10%	DB	0.7	1.2	2.7	7.3	16.3	19.7	23.5	20.5	15.7	10.2	6.5	3.3	(25)
(26)			MCWB	0.4	0.8	1.1	4.2	11.2	14.9	18.5	16.5	13.6	9.0	5.9	2.7	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.1	3.2	4.7	10.2	16.0	19.6	21.8	21.7	16.6	11.8	9.2	7.2	(27)
(28)			MCDWB	3.5	4.6	6.6	12.6	19.8	23.9	26.3	25.8	18.8	12.5	10.2	7.9	(28)
(29)		2%	WB	2.1	2.3	3.1	7.4	14.6	17.6	20.9	19.6	15.7	11.1	8.2	4.7	(29)
(30)			MCDWB	2.4	3.3	4.8	11.3	18.6	21.4	24.9	22.8	17.3	11.8	8.9	5.2	(30)
(31)		5%	WB	1.2	1.6	2.2	6.0	13.4	16.5	20.2	18.3	14.9	10.3	7.1	3.6	(31)
(32)			MCDWB	1.4	2.1	3.4	8.9	17.5	20.2	24.0	21.1	16.4	11.0	7.6	4.0	(32)
(33)		10%	WB	0.5	0.9	1.4	4.6	11.7	15.5	19.3	17.2	14.1	9.1	6.0	2.7	(33)
(34)			MCDWB	0.8	1.2	2.4	6.8	16.0	19.1	22.8	19.8	15.5	10.0	6.5	3.2	(34)
(35)	Mean Daily Temperature Range	MDBR		4.7	5.0	6.5	7.2	7.8	6.4	6.1	5.3	4.4	3.5	3.0	3.3	(35)
(36)		5% DB	MCDBR	3.3	4.3	7.1	10.6	11.2	9.2	8.1	7.2	5.8	4.1	3.8	3.7	(36)
(37)			MCWBR	3.1	3.6	5.0	6.1	5.3	4.0	3.6	3.7	3.5	3.3	3.6	3.6	(37)
(38)		5% WB	MCDWB	3.2	4.2	5.8	9.4	10.4	7.9	7.2	6.2	5.2	3.9	3.8	3.7	(38)
(39)			MCWBR	3.1	3.7	4.5	6.0	5.5	3.8	3.7	3.8	3.4	3.7	3.6	3.6	(39)
(40)	Clear-Sky Solar Irradiance	taub		0.247	0.292	0.320	0.359	0.356	0.352	0.376	0.361	0.338	0.319	0.269	0.357	(40)
(41)		taud		2.064	2.147	2.228	2.353	2.435	2.475	2.425	2.456	2.512	2.412	2.162	2.600	(41)
(42)		Ebn at Noon		442	661	786	831	865	875	840	823	780	647	414	290	(42)
(43)		Edh at Noon		37	69	92	100	100	98	100	90	71	55	34	23	(43)
(44)	All-Sky Solar Radiation	RadAvg		0.22	0.78	2.25	3.67	4.95	5.22	5.03	3.90	2.27	0.95	0.25	0.12	(44)
(45)		RadStd		0.04	0.18	0.28	0.40	0.55	0.41	0.51	0.45	0.30	0.17	0.05	0.01	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (56 neighbors)		+0.49	N/A	N/A	N/A	N/A	N/A	-148	N/A	N/A	N/A

Nomenclature: See separate page